

Präsentation

Teil 3.0- 3.2

Präsentiert von:

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Gliederung Präsentation:

3. Power Window Controller Interactive Testing and
Experimental Results

3.1 Case 1: Window Up. Case 2: Window Auto Up

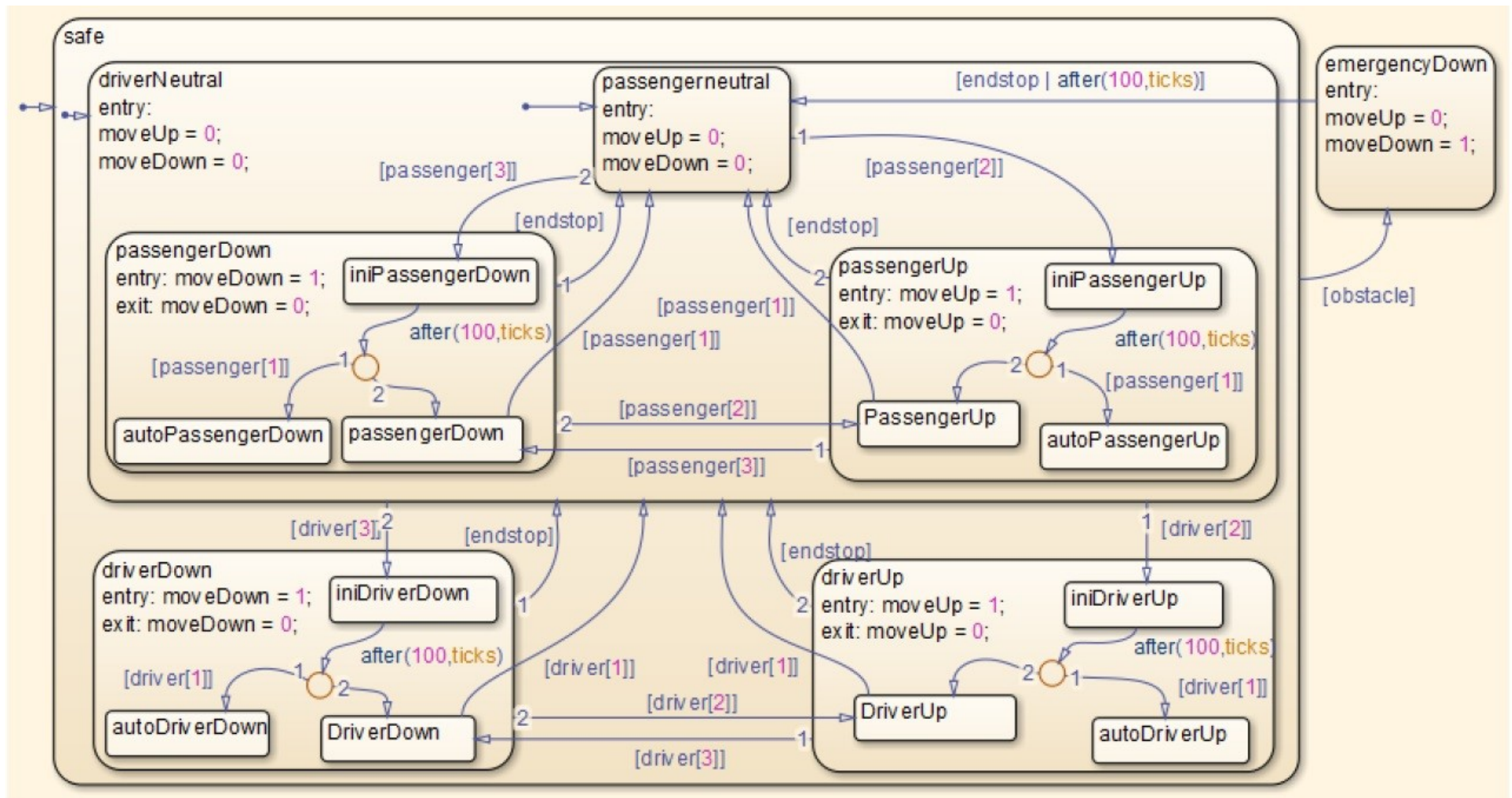
3.2 Case 3: Driver Side Precedence

Präsentation Teil 1

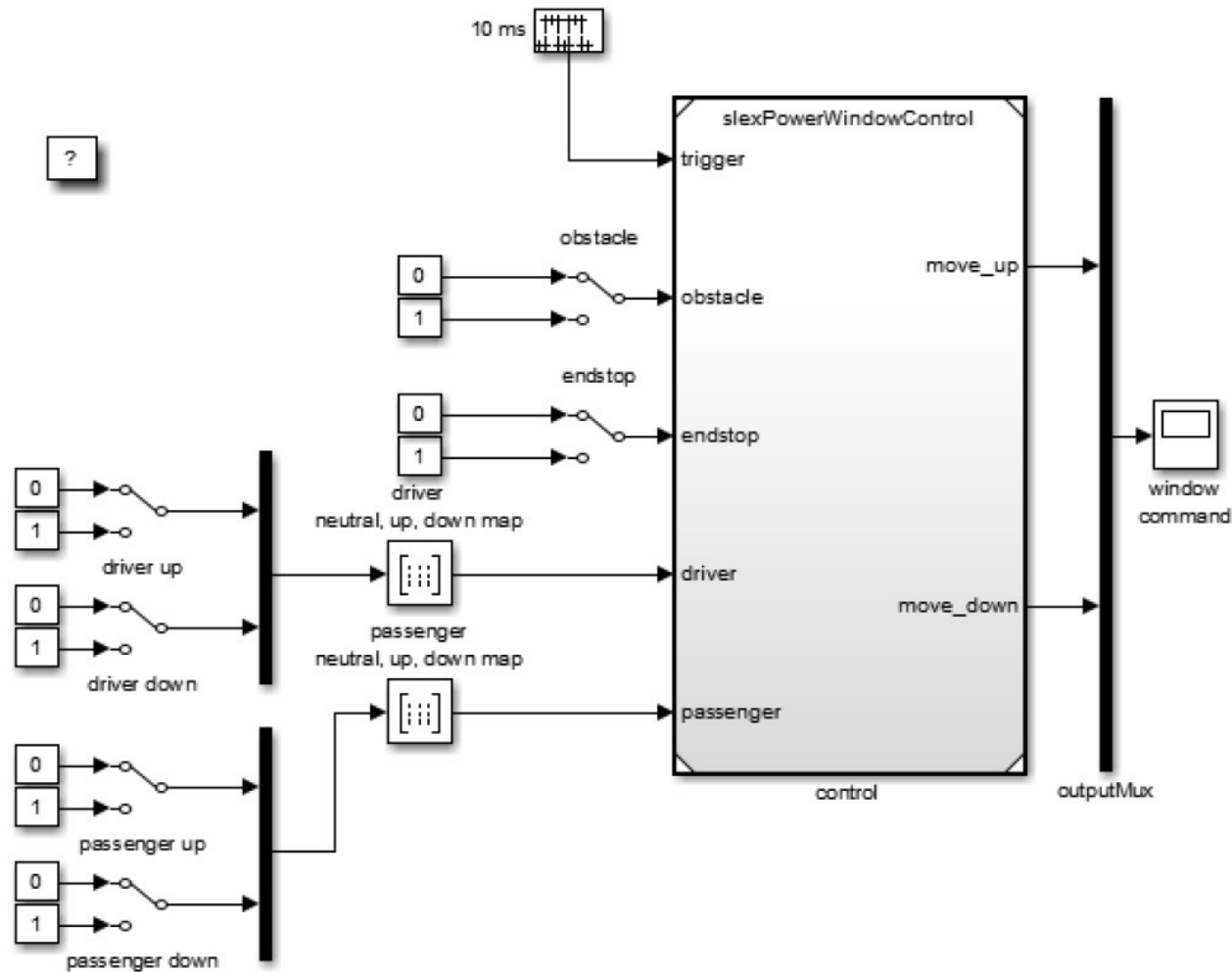
3. Power Window Controller Interactive Testing and Experimental Results

by Andreas Hiebert

Interactive Testing



Simulink Modell



Passenger Input

Element	Description
neutral	Passenger control switch is not depressed.
up	Passenger control switch generates the up signal.
down	Passenger control switch generates the down signal.

Driver Input

Element	Description
neutral	Driver control switch is not depressed.
up	Driver control switch generates the up signal.
down	Driver control switch generates the down signal.

Window Up or Down

Element	Description
0	Window moves freely between top or bottom.
1	Window is stuck at the top or bottom because of physical limitations.

Obstacle in Window

Element	Description
0	Window moves freely between top or bottom.
1	Window has obstacle in the frame.

Inputs		Outputs		
up	down	up	down	neutral
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

Präsentation Teil 2

3.1

Case 1 : Window UP

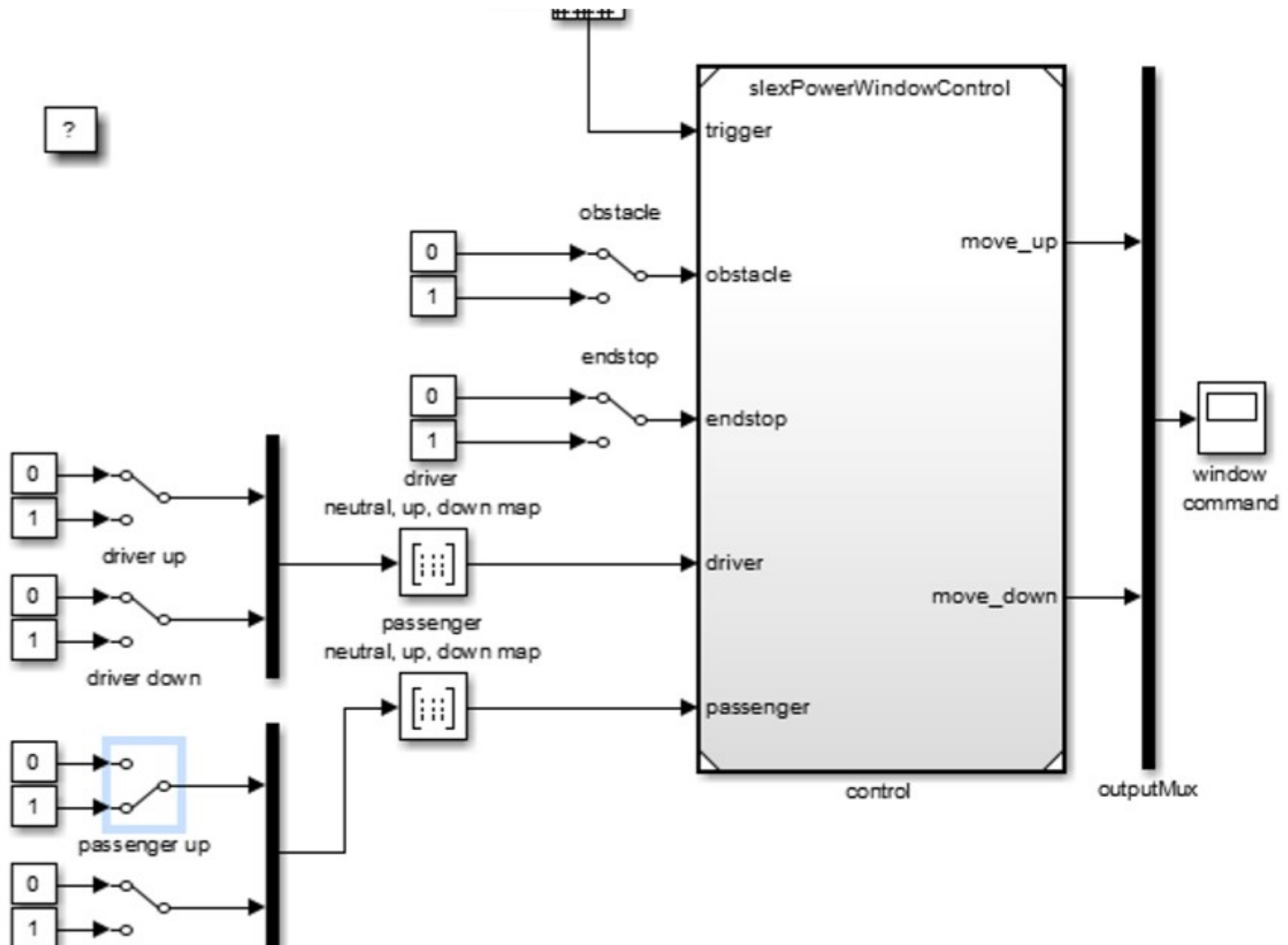
Case 2: Window Auto-UP

Case 3 : Driver-Side Precedence

By Meem Miah

- Open the `slexPowerWindowCntlInteract` model

Case 1



Case 1

- What happens if the passenger presses the PHYSICAL WINDOW SWITCH for more than one second?



Case 1 : Window Up

->window moves up

Or

->Endstop Event is generated

Case 1 : Window Up

- What is „Endstop Event“ and when will it be generated?



Case 2

- What happens if the passenger presses the PHYSICAL WINDOW SWITCH for less than one second?



Case 2 : Window Auto-UP

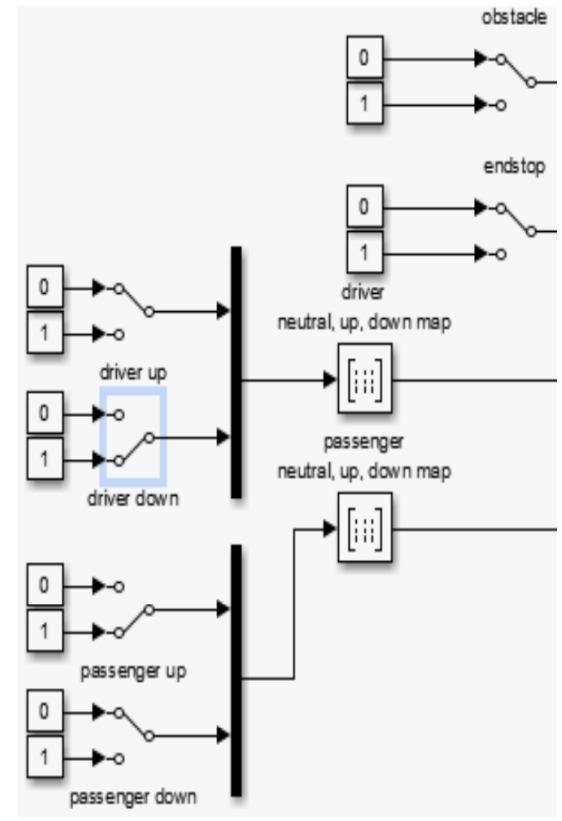
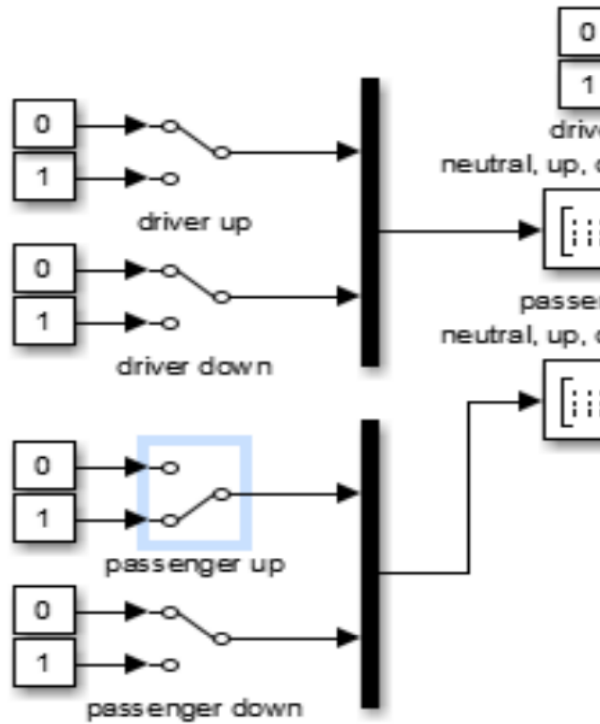
->Auto Up behaviour will be activated

Case 3

- What happens if both the driver and the passenger want to control the same passenger window (in opposite directions) at the same time?



Case 3



Case 3 : Driver-Side Precedence

Example :

Passenger-Up = 1

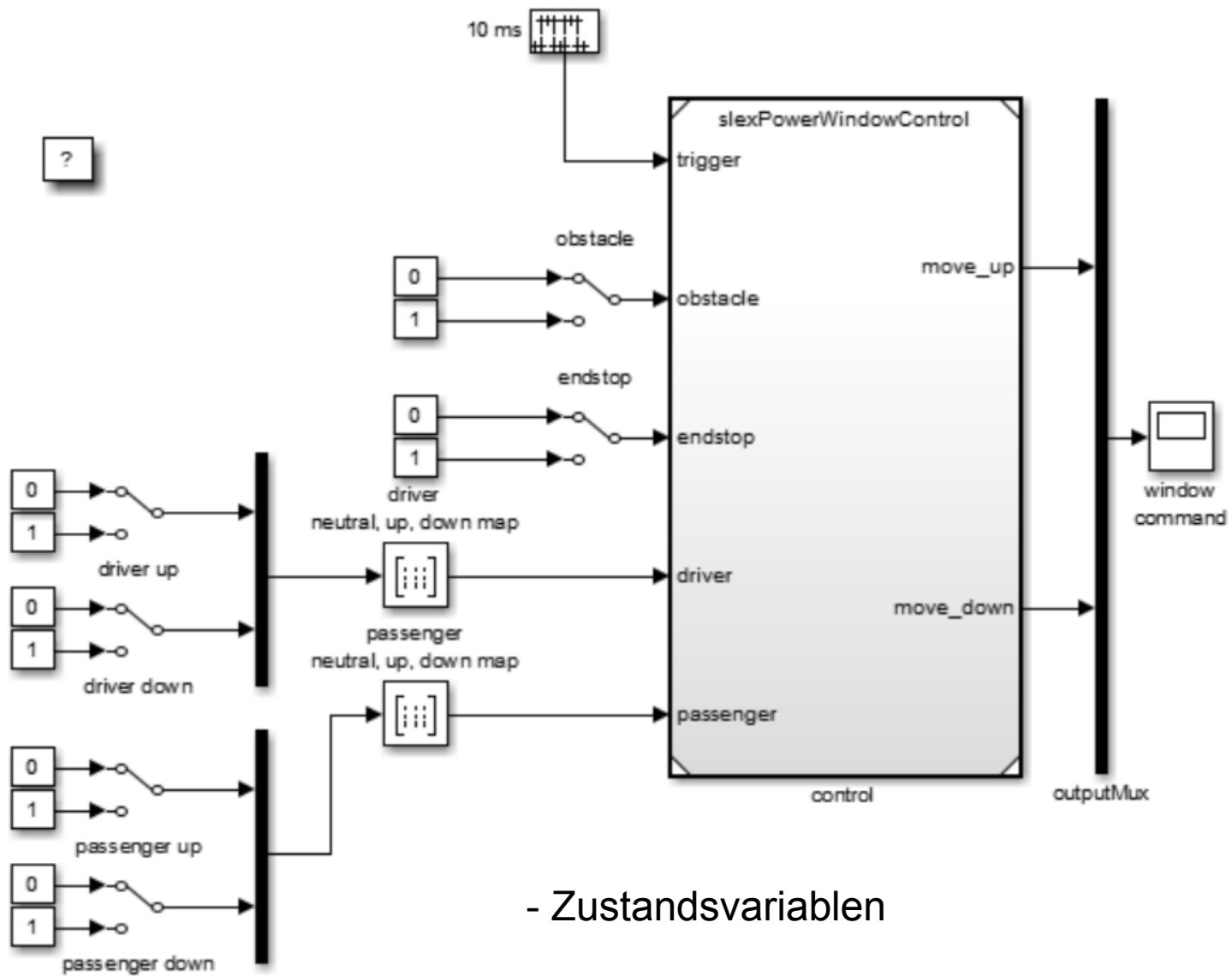
Driver-Down = 1

→ Window Down Output will be generated

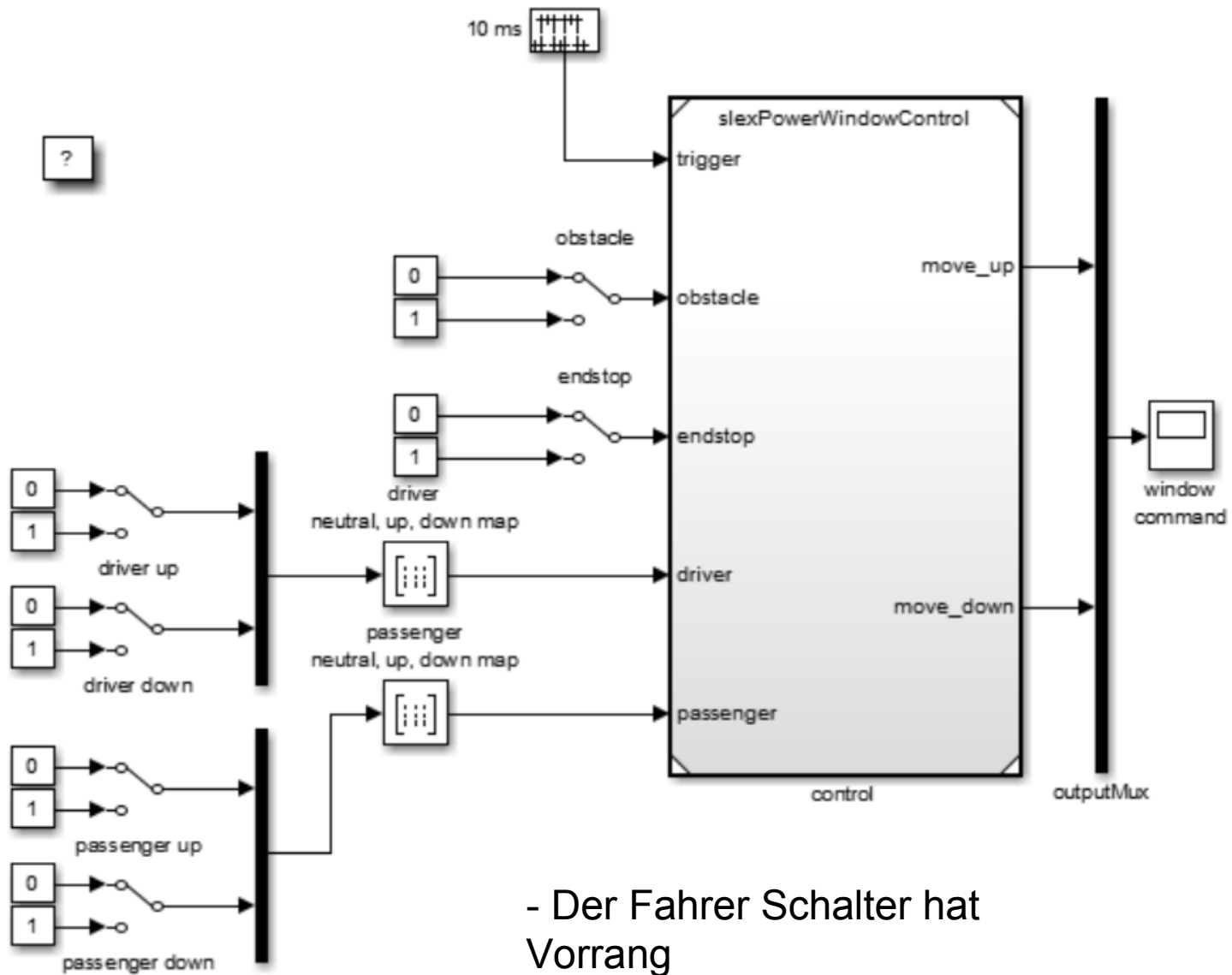
Präsentation Teil 3

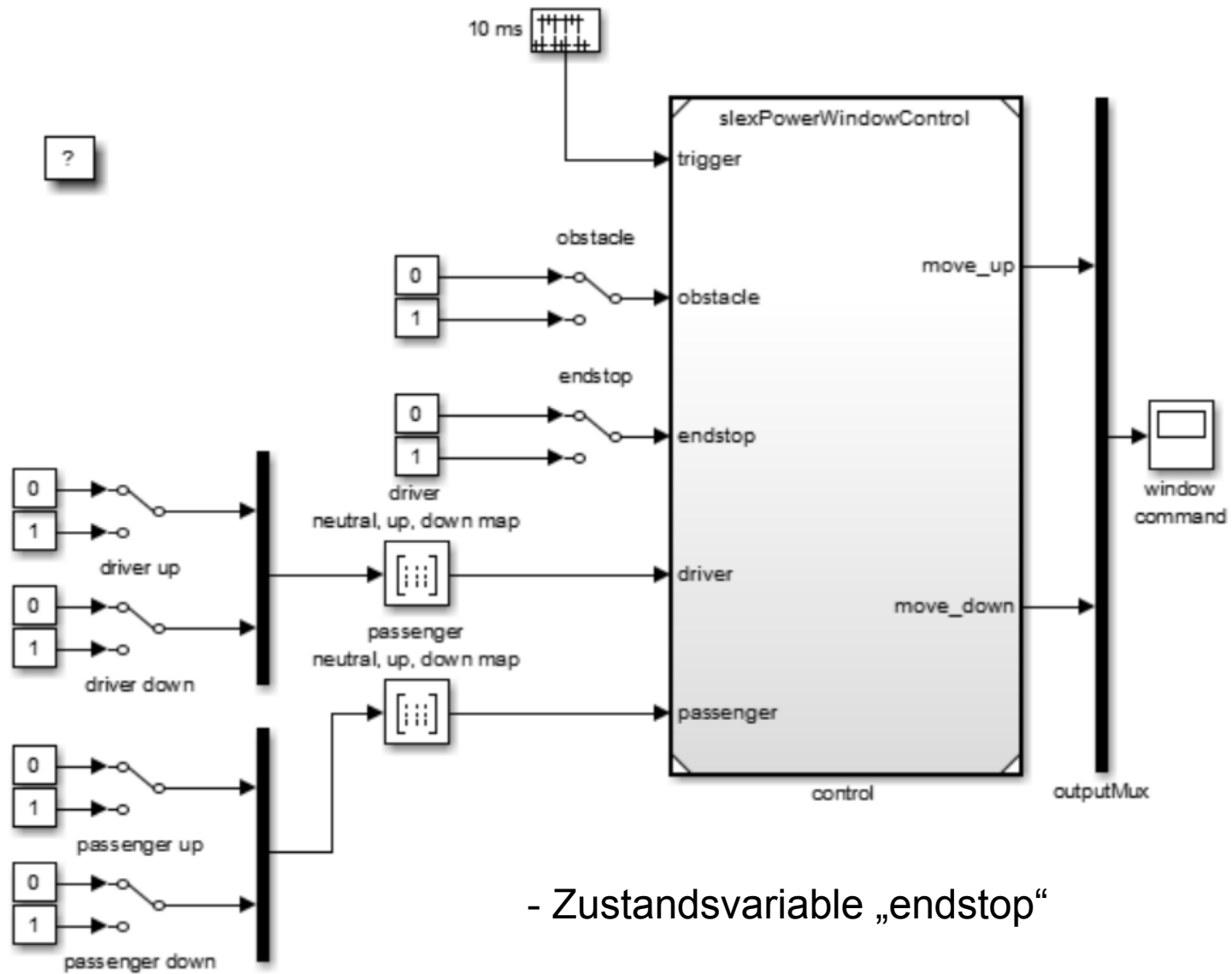
3.2 Case 3: Driver Side Precedence

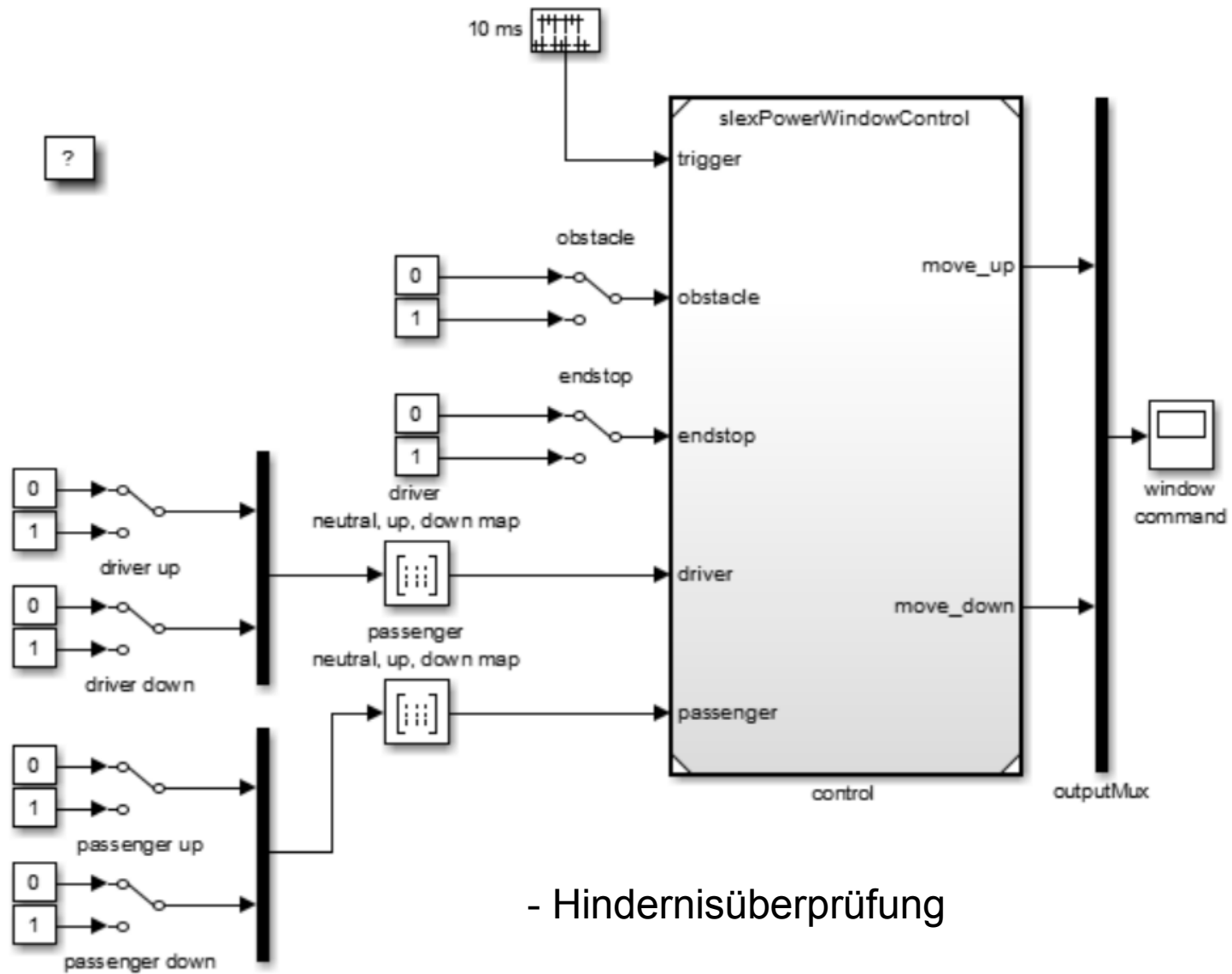
by Jens Ramermann



- Zustandsvariablen







- Hindernisüberprüfung

Vielen Dank für Eure
Aufmerksamkeit